DEVOPS ENGINEERING - UNIT 2 MINI PROJECT

Enterprise Student Record and Academic Management Platform

Agile Product Planning, Scrum Framework, Backlog & Release Engineering

Student Name: Kartik Supekar | **PRN:** 2125UDSM1121

Course / Unit: DevOps (Unit 2 Mini Project) | **Submission Date:** September 04, 2026

1. Review of Business Problem & Product Discovery (Unit 1 Alignment)

In Unit 1, an extensive Design Thinking and domain discovery process was conducted to understand the operational complexities of higher education institutions. The Enterprise Student Record and Academic Management Platform (EduTrack Enterprise) addresses the historical inefficiencies of manual academic paper registers, disjointed spreadsheets, and bureaucratic grade compilation delays.

Core Takeaways from Unit 1 Discovery:

- **Product Vision Alignment:** Develop a resilient, cloud-native academic management system that ensures data integrity and eliminates repetitive administrative workload.
- **Target Users & Personas:** Directly serves Students (real-time progress, attendance tracking), Faculty (speedy attendance, grade entries), and Exam Administrators (result compilation, verification, lock-in).
- **Minimum Viable Product (MVP) Scope:** Focuses on Authentication & RBAC, Student Profiles, Course Enrollment, Daily Attendance Tracking, and SGPA/CGPA Grade Card generation.
- **DevOps Progression:** Unit 1 established the 'Why' and 'What' (Design Thinking and Architecture). Unit 2 operationalizes the 'How' using Agile Scrum frameworks to plan sprint cadences and incremental deliverables.

2. Agile Product Vision & Project Capabilities

Agile Product Vision Statement: *"Develop a secure, intuitive, and highly scalable Student Record and Academic Management Platform that empowers higher education institutions to seamlessly coordinate student onboarding, course registrations, daily attendance audits, examination grading, and verifiable reporting through modern, container-ready cloud workflows."*

2.1 Agile Vision Dimensions & Capabilities Matrix

Dimension	Agile Target Objective	Measurable Success Benchmark
Primary Users	Students, Faculty Members, Academic Administrators, Exam Cell Staff	100% adoption across campus departments
Core Business Benefits	Paperless record handling, real-time attendance alerts, automated transcript generation	Administrative overhead reduced by 85%; zero lost paper records
Measurable Success Criteria	Sub-second API response time (<300ms), 99.9% platform availability, instant GPA computation	Result processing turnaround slashed from 20 days to 24 hours
Expected Capabilities	Session attendance marker, prerequisite credit checking, PDF grade cards, RBAC security	Zero enrollment prerequisite errors; automated audit trails
Project Constraints	Fixed 2-week Sprint cadences, relational integrity constraints (PostgreSQL), zero production downtime	Strict adherence to DoD and automated CI/CD gating

3. Scrum Team Organization & Role Responsibilities

To ensure disciplined Agile execution, the project structure is organized around core Scrum roles with cross-functional accountability:

Scrum Role	Designated Lead / Member	Core Responsibilities	Key Artifacts Owned
Product Owner (PO)	Kartik Supekar *(Lead PO)*	• Defines vision & product requirements • Prioritizes the Product Backlog • Represents university stakeholder interests • Accepts/rejects completed sprint stories	Product Backlog, User Story Definitions, Acceptance Criteria
Scrum Master (SM)	Kartik Supekar *(Agile Facilitator)*	• Facilitates Scrum ceremonies • Eliminates technical & resource impediments • Protects team from scope creep • Enforces Agile & Definition of Done guidelines	Sprint Burndown Charts, Impediment Log, Velocity Tracker
Development Team	Cross-Functional Engineering Squad *(Frontend, Backend, DevOps, QA)*	• Technical design & schema implementation • Writing clean, tested application code • Story point estimation via Planning Poker • Automated CI/CD pipelines & test suites	Sprint Backlog, Potentially Shippable Increment, Test Reports

4. Project Epics (High-Level Functional Modules)

The overall platform architecture is broken down into 7 high-level business modules known as Epics, covering the entire educational management lifecycle:

Epic ID	Module Title & Scope	Encompassed Functional Features
EPIC-01	Student Management Module	Student Registration, PRN Generation, Profile Management, Search & Multi-Filter Query, Profile Updates, Student Archival/Deletion
EPIC-02	Course Management Module	Course Catalog Publishing, Add Course, Update Course Syllabus/ Credits, Assign Students to Batches, View Course Enrollment Lists
EPIC-03	Attendance Management Module	Session-Based Lecture Attendance Marking, Fast-Grid Selection, View Aggregate Percentages, Automated Attendance Status Reports
EPIC-04	Marks & Academic Performance Module	CIA & End-Sem Marks Entry, Score Updates, Automated SGPA/CGPA Grade Calculation Engine, Verified Result Sheet Generation
EPIC-05	Authentication & Security Module	User Login, Session Management, Logout, Password Reset, JWT-based Role-Based Access Control (RBAC: Student, Faculty, Admin, Exam Cell)
EPIC-06	Notifications & Alerting Module	Low-Attendance Trigger Alerts (<75%), Examination Hall Ticket Notifications, Semester Result Publication Broadcasts
EPIC-07	Reporting & Executive Analytics Module	Student Academic Records, Consolidated Attendance Reports, Departmental Performance Summaries, PDF Transcript Exporters

5. User Story Repository & Acceptance Criteria (Definition of Done)

Each Epic is decomposed into granular User Stories following the standard industry template: *'As a <Role>, I want <Feature>, so that <Benefit>'*. Each story contains explicit, testable Acceptance Criteria:

> **US-01 (Authentication)**
>
> *User Story:* As a Campus User (Student/Faculty/Admin), I want to log in using my university credentials so that I can securely access my role-specific dashboard.
>
> **Acceptance Criteria (Definition of Done):**
>
> - Login interface loads securely under HTTPS with email and password inputs.
> - System validates credentials against Bcrypt password hashes in the database.
> - Invalid login attempts trigger a clean, descriptive error message without leaking sensitive data.
> - Successful authentication issues a role-specific JWT session token and redirects to the appropriate dashboard.

US-02 (Student Management)

User Story: As an Academic Administrator, I want to register a new student with personal details and course batch so that a unique PRN is allocated.

Acceptance Criteria (Definition of Done):

- Admin form accepts full name, date of birth, contact email, branch, and academic intake year.
- System auto-generates a unique permanent registration number (PRN: 2125UDSM1121).
- Record is committed to PostgreSQL with immediate visual confirmation.
- Duplicate registrations against same national ID or email are rejected with validation errors.

US-03 (Course Management)

User Story: As an Academic Coordinator, I want to create a course with credit allocations and prerequisites so that students can register for it.

Acceptance Criteria (Definition of Done):

- Interface provides inputs for Course Code, Title, Department, Credits (1-5), and Prerequisite Course IDs.
- Course code is enforced as unique across the academic catalog.
- New courses immediately become available in the elective registration window.

US-04 (Attendance Tracking)

User Story: As a Faculty Member, I want to record attendance for my assigned lecture session so that attendance ledgers remain current and audit-compliant.

Acceptance Criteria (Definition of Done):

- Faculty selects academic course, section, and lecture date.
- Student roster displays with a single-click 'Mark All Present' toggle.
- Attendance records persist immediately upon submission with a timestamp.
- System calculates aggregate student attendance percentages dynamically.

US-05 (Performance & Grading)

User Story: As a Faculty Member, I want to enter internal assessment and exam marks for students so that semester grades are calculated accurately.

Acceptance Criteria (Definition of Done):

- Marks entry grid validates input numbers against maximum evaluation caps (e.g., 0-30 for CIA).
- System prevents out-of-range numerical submissions with inline warnings.
- Grading engine calculates SGPA and letter grade (A+, A, B, etc.) based on standard institutional curves.
- Marks are marked as 'Draft' until submitted to the Exam Cell for moderation.

US-06 (Notifications & Alerts)

User Story: As a Student, I want to receive an automated notification when my attendance drops below 75% so that I can take corrective action before exams.

Acceptance Criteria (Definition of Done):

- Background cron job audits cumulative attendance weekly.
- Trigger fires automated email and in-app dashboard warning for students <75%.
- Notification specifies the exact deficit hours required to regain eligibility.

US-07 (Reporting & Transcripts)

User Story: As an Examination Controller, I want to generate verified PDF grade cards with one click so that students receive their semester transcripts promptly.

Acceptance Criteria (Definition of Done):

- System compiles all verified course grades, SGPA, CGPA, and institutional stamp.
- Outputs a secure, formatted PDF document with cryptographic verification hash.
- Batch export functionality allows downloading grade cards for an entire department.

6. Prioritized Product Backlog & Story Point Estimation

Backlog prioritization was conducted using the MOSCOW framework and Planning Poker relative sizing based on technical complexity, dependencies, and stakeholder business value:

Planning Poker Fibonacci Scale: 1 (Very Easy) | 2 (Easy) | 3 (Moderate) | 5 (Complex) | 8 (Very Complex) | 13 (Highly Architectural)

Story ID	Target Epic	User Story Summary	Priority	Complexity	Points
US-01	Authentication	User Login, JWT Generation & RBAC Middleware	High (Must)	Complex	5
US-02	Student Mgmt	Student Admission Form & PRN Allocation	High (Must)	Moderate	3
US-03	Student Mgmt	Multi-criteria Student Search & Profile Filtering	High (Must)	Easy	2
US-04	Student Mgmt	Student Profile Update & Document Verification	Medium (Should)	Moderate	3
US-05	Course Mgmt	Add New Course & Credit Ceiling Enforcement	High (Must)	Moderate	3
US-06	Course Mgmt	Student Course Registration & Prerequisite Checker	High (Must)	Complex	5
US-07	Attendance	Faculty Lecture Attendance Marker & Bulk Toggle	High (Must)	Complex	5
US-08	Attendance	Student Self-Service Attendance Dashboard	Medium (Should)	Easy	2
US-09	Marks & Grades	Faculty Assessment Marks Entry & Auto-Grade Calc	High (Must)	Complex	5
US-10	Reporting	1-Click PDF Semester Grade Card Generation	High (Must)	Complex	5

7. Sprint Planning & Multi-Sprint Cadence

The development phase is organized into three balanced 2-week Sprints, ensuring an incremental progression from core architecture to full reporting:

Sprint	Sprint Goal	Allocated User Stories & Activities	Committed Velocity
Sprint 1 *(Weeks 1-2)*	Establish Secure Architecture, Authentication & Core Student Onboarding	• Git repository & Docker environment setup • US-01: User Login, JWT Auth & RBAC (5 pts) • US-02: Student Admission & PRN Generator (3 pts) • US-03: Student Search & Filter API (2 pts) • Automated Jest Unit Tests & CI Pipeline	15 Story Points
Sprint 2 *(Weeks 3-4)*	Implement Course Registration & Session-Based Attendance Engine	• US-04: Student Profile Updates (3 pts) • US-05: Course Catalog & Credit Limits (3 pts) • US-06: Course Registration & Prerequisite	18 Story Points

Sprint	Sprint Goal	Allocated User Stories & Activities	Committed Velocity
		Validation (5 pts) • US-07: Faculty Attendance Marker (5 pts) • US-08: Student Attendance Dashboard (2 pts)	
Sprint 3 *(Weeks 5-6)*	Deliver Examination Grading, Alerts & Verifiable PDF Reporting	• US-09: Marks Entry & SGPA/CGPA Calculation (5 pts) • US-10: 1-Click PDF Grade Card Exporter (5 pts) • Notifications: Low attendance warning alerts (3 pts) • End-to-end integration & load testing	13 Story Points

8. Scrum Board Execution & Progress Visualization

A dynamic 5-column Scrum Board is maintained to visualize work in progress, identify task blockers, and enforce WIP limits across the sprint.

9. Sprint Burndown Chart & Ceremony Simulation

Progress tracking throughout Sprint 1 was quantified using a 10-day sprint burndown tracking 34 initial story points down to completion.

9.1 Simulation of Four Mandatory Scrum Ceremonies

- **1. Sprint Planning Meeting:** Product Owner presented user stories from the top of the backlog. The team estimated effort via Planning Poker, agreed on a Sprint Goal of 'Delivering Secure Auth & Basic Student Onboarding', and committed to 15 Story Points for Sprint 1.
- **2. Daily Scrum (Stand-Up):** Daily 15-minute sync where each engineer answered: (1) What did I complete yesterday? (2) What will I work on today? (3) What impediments are blocking my progress? Example impediment: Database migration lock resolved by DBA within 2 hours.
- **3. Sprint Review:** Demonstrated live working software increment to academic stakeholders (Registrar and Faculty leads). Walked through successful student login, role-based page protection, and instant student admission with PRN allocation. Stakeholders approved the increment.
- **4. Sprint Retrospective:** Analyzed sprint dynamics: What went well (automated Docker container builds saved onboarding time); What didn't go well (initial confusion regarding prerequisite credit calculation); Action item (write explicit API schema contracts before frontend development begins).

10. Comparative Study of Agile Frameworks & Justification

A comprehensive evaluation of leading Agile frameworks was conducted to identify the most suitable development methodology for the academic platform:

Framework	Core Philosophy	Key Strengths	Target Project Fit
Scrum	Time-boxed iterations (Sprints), fixed roles (PO/SM/Dev), explicit ceremonies.	Predictable cadences, rapid feedback loops, transparent accountability.	Small to medium cross-functional product teams (EduTrack Platform).
Kanban	Continuous flow, visual WIP limits, pull-system without fixed time-boxes.	Exceptional flexibility, ideal for maintenance and operations tickets.	DevOps infrastructure operations & ongoing bug-fix maintenance.
Extreme Programming (XP)	Engineering-centric: TDD, pair programming, CI, refactoring.	Very high code quality, automated test coverage, reduced technical debt.	High-risk algorithmic and grading engine development modules.

Framework	Core Philosophy	Key Strengths	Target Project Fit
Lean Development	Eliminate waste, amplify learning, deliver as fast as possible.	Resource efficiency, cost minimization, value-stream focus.	Early product startup discovery and MVP feature trimming.
Large Scale Scrum (LeSS) / SAFe	Single Product Backlog scaled across multiple teams (2-8 teams); enterprise governance.	Maintains simplicity of Scrum while scaling across multiple teams.	Medium multi-department systems to massive university networks.
Scrum@Scale	Modular scaling via Scrum of Scrums and Executive MetaScrum.	Decentralized coordination with minimal bureaucratic overhead.	Fast-growing educational software companies with expanding teams.

10.1 Selected Framework Justification

Recommendation: Agile Scrum with Kanban Hybrid (Scrumban)

For the Student Record and Academic Management Platform, Scrum is selected as the primary delivery framework. Its structured 2-week time-boxed sprints guarantee predictable deliverables for academic stakeholders, while daily stand-ups and retrospective feedback loops protect the team against scope creep. For ongoing DevOps infrastructure and deployment tasks, a Kanban WIP board is integrated to handle real-time server and CI/CD operations.

11. Deliverables Verification & Sign-Off Checklist

All student deliverables specified in the Unit 2 Agile Mini-Project Curriculum have been executed and documented:

- ✔ Agile Project Planning Report & Business Domain Review (Completed)
- ✔ Agile Product Vision Document & Measurable Success Metrics (Completed)
- ✔ Scrum Team Structure & Role Accountability Specification (Completed)
- ✔ Epic Documentation covering 7 full academic management modules (Completed)
- ✔ User Story Repository with detailed Definition of Done / Acceptance Criteria (Completed)
- ✔ Prioritized Product Backlog using MOSCOW and Planning Poker Story Points (Completed)
- ✔ Sprint Planning Document with 3 Phased 2-Week Sprints (Completed)
- ✔ Sprint Backlog with committed story point velocity (Completed)
- ✔ Scrum Board Diagram illustrating active task status and WIP flow (Completed)
- ✔ Sprint 1 Burndown Chart quantifying actual vs. ideal progress (Completed)
- ✔ Scrum Ceremonies Simulation Report (Planning, Daily Scrum, Review, Retrospective) (Completed)
- ✔ Comprehensive Comparative Study of 7 Agile Frameworks with Technical Justification (Completed)

Submission Sign-Off:

Student Name: Kartik Supekar | PRN: 2125UDSM1121 | Date: September 04, 2026
Course: DevOps Engineering (Unit 2 Mini Project)